

Workspace Manager — Capstone Project

Frontend-only feature list | Next.js + Tailwind + Redux Toolkit | Notion/Jira-style workspace tool

This document lists every feature to implement for the capstone project, grouped into 14 domains (70 discrete features total). No backend is used — auth, real-time updates, and persistence are all simulated on the client using local/mock data, localStorage, and IndexedDB.

Auth & User (mocked, no real backend)

- Login/signup screen with fake credential check against local mock users
- Persisted “logged in” session via localStorage
- User profile with avatar, name, email (editable, stored locally)
- Multiple mock user profiles to switch between (simulate multi-user)
- Logout clears session state

Workspaces

- Create/rename/delete workspace
- Switch between multiple workspaces
- Workspace-level settings (name, icon/color, default view)
- Workspace member list (mock users assigned to it)
- Invite member (fake — just adds a mock user to the list)

Projects

- Create/rename/archive/delete project within a workspace
- Project color/icon tagging
- Project description field
- Project-level member assignment (subset of workspace members)
- Project templates (predefined task sets to start from)

Tasks & Subtasks

- Create/edit/delete task
- Nested subtasks (checklist-style, arbitrary depth or 1-level)
- Task fields: title, description, status, priority, due date, assignee, labels/tags
- Mark task/subtask complete
- Task detail modal/page (expanded view)
- Attach files to a task (stored as base64/blob in state, not uploaded anywhere)
- Convert subtask to full task and vice versa
- Duplicate task
- Bulk actions (multi-select → change status/assignee/delete)

Views (same data, multiple representations)

- Kanban board view with drag-and-drop between columns
- Custom/reorderable kanban columns per project
- List/table view with sortable columns
- Calendar view showing tasks by due date
- Toggle/persist last-used view per project
- Group-by option in list view (by assignee, status, priority, label)

Filtering, Search, Sort

- Global search across tasks/projects/workspaces
- Filter by assignee, label, priority, status, due date range
- Sort by due date, priority, created date, alphabetical
- Saved filter presets

Roles & Permissions (simulated client-side)

- Role types: owner, admin, member, viewer
- Permission-gated UI (e.g. viewer can't edit/delete)
- Role assignment per workspace member
- "Access denied" states for restricted actions

Activity Log

- Per-task activity feed (created, edited, status changed, commented)
- Per-project activity feed aggregating task events
- Timestamped entries with acting user
- Filterable activity log (by user, by action type)

Comments & Collaboration (simulated, no real backend)

- Comment thread per task
- @mention autocomplete (mock users)
- Edit/delete own comments
- Simulated "live" updates via setInterval/mock socket pushing fake events

Undo/Redo & Optimistic UX

- Undo/redo stack for task edits, moves, deletes
- Optimistic UI updates with simulated network delay + rollback on fake failure
- Toast notifications for actions (with undo action inline, e.g. "Task deleted — Undo")

Notifications (in-app only)

- Notification bell with unread count
- Notification triggers: assigned to task, mentioned in comment, due date approaching
- Mark as read / mark all as read

- Notification preferences toggle (per event type)

Data Persistence & Offline

- Full app state persisted to localStorage/IndexedDB
- Rehydrate state on reload
- “Offline” indicator (simulate via navigator.onLine)
- Manual “sync” button that fakes reconciling offline changes
- Export workspace data as JSON
- Import workspace data from JSON (with validation)
- Reset/clear all data option

UI/UX Utilities

- Command palette (Cmd+K) for quick navigation/actions
- Keyboard shortcuts (create task, navigate views, etc.)
- Dark/light theme toggle (persisted)
- Responsive layout (mobile drawer nav, collapsed sidebar)
- Loading skeletons for simulated async operations
- Empty states for no tasks/projects/search results
- Confirmation dialogs for destructive actions

Settings

- App-wide preferences (theme, default view, notification toggles)
- Workspace settings page
- Danger zone (delete workspace/project with confirmation)